

ICE: Intercede Concept Erasure in Text-to-Image Diffusion Models

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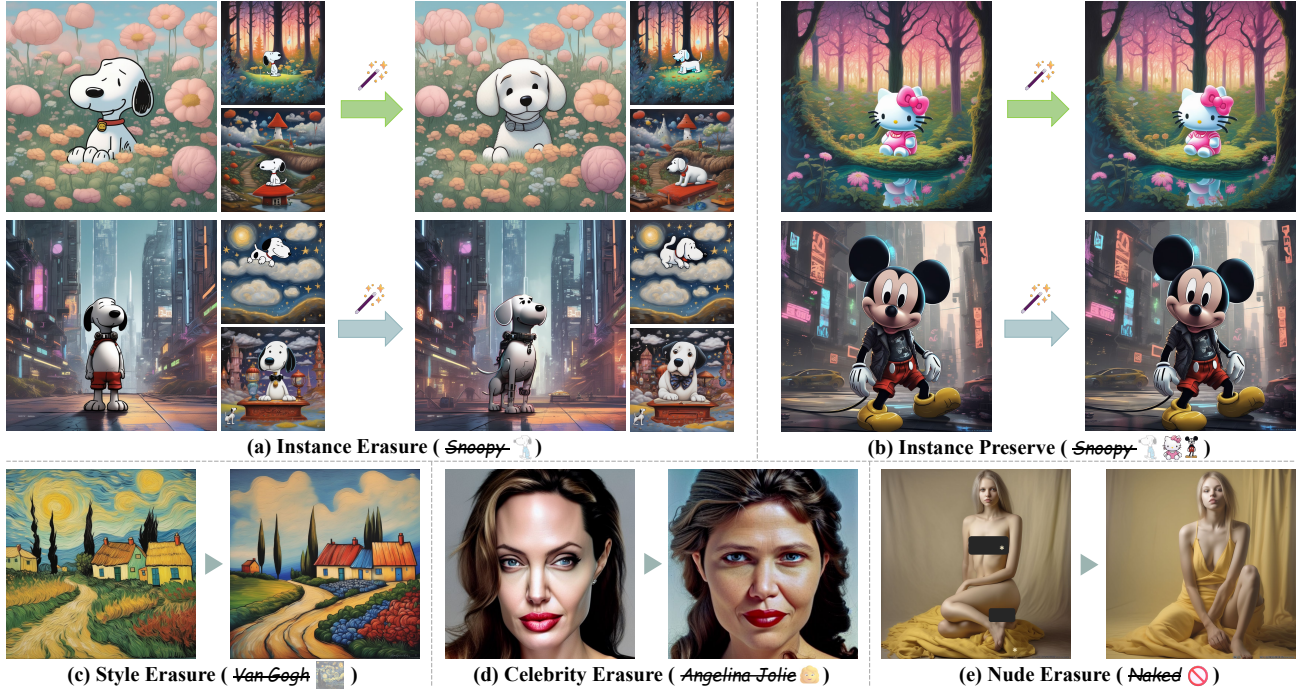


Figure 1: Our proposed method, ICE, can erase specified concepts from text-to-image diffusion models. This can protect (a) the copyright of cartoon characters, (c) the copyright of artworks, (d) the right of publicity of celebrities, and (e) explicit content; while fully preserving (b) non-target concepts.

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ABSTRACT

The success of diffusion models in text-to-image (T2I) generation has made it urgent to remove unwanted concepts, such as copyrighted, offensive, and unsafe ones, from pre-trained models in an accurate, timely, and cost-effective manner. However, limited by the inherent optimization perspective, existing methods have

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two major problems. Firstly, they overlook maintaining the global visual style during the erasure process, leading to significant style shifts. Secondly, excessive concept erasure causes relevant content to disappear or generates substitutes unrelated to the original object's attributes. Compared to other methods, our proposed ICE has unique advantages, as it can generate diverse visual features and achieve a balance between concept erasure and maintaining the semantic content of the target object. This is mainly achieved through our well-designed non-erasable features protector (NEFP) and augmented invariant constraints (AIC). Specifically, we enhance the protection of feature information by embedding an augmented orthogonal anchor concept matrix. Meanwhile, under controlled constraints, we introduce invariants into the embedding space to retain key semantics. This work specifically emphasizes the importance of focusing on feature expression and semantic protection in the concept erasure task for fully unleashing the performance of T2I models.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision**.

KEYWORDS

Generation safety, concept erasure, T2I model, anchor matrix

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1 INTRODUCTION

The rapid development of large-scale text-to-image(T2I) models [15, 31, 36, 46] has enabled them to generate high-quality and realistic images, thus gaining widespread multimedia application. However, these models also raise a series of issues, including portrait infringement [2, 14, 22, 44], copyright infringement [4, 8, 10, 13, 17, 21, 38], and the generation of explicit information [16, 25, 37, 42]. While the most straightforward solution [34] is to clean the data and retrain the large-scale generative models, this approach is relatively costly and inefficient [30]. Therefore, it is particularly important to develop a low-cost technique that can accurately erase unnecessary semantic concepts from images, known as concept erasure [6, 7, 20, 24, 26, 45]. The task of concept erasure involves precisely removing the visual content of target concepts from generated images while preserving the non-target concepts, ensuring that the diffusion model can safely perform T2I generation.

Recent methods have focused on fine-tuning, text guidance adjustment, and targeted intervention in the generation process. Although these efforts have effectively erased target concepts, they often come at the expense of sacrificing non-erased content. We identify three main issues that hinder the effectiveness of previous work. First, erasing the main concept in a styled image, influenced by the T2I model's prior data information and the entanglement of "snoopy" with its comic style, can cause a global style shift in the resulting image, as shown in Figure 2(a). Second, if the text prompt



Figure 2: We have observed three key issues in concept erasure in the past: (a) Erasing the target concept also affects the overall style presentation. (b) After erasing “snoopy,” the main subject becomes a “boat” instead of a “dog,” showing a lack of understanding of the target semantics. (c) The target content is completely removed, resulting in over-erasure that is inconsistent with user intentions.

only contains erasure information such as “~~snoopy~~” without specifying the replaced main concept, it may randomly generate any item. As shown in Figure 2(b), a boat is generated instead of a dog. The noise addition process retains only a blurred outline of the image, and as the denoising process progresses, the result will only proceed along the wrong sampling direction. Another common problem is shown in Figure 2(c), where the erasure process causes the main content within the image to disappear completely. The definition of concept erasure [6, 42] is to prevent the model from generating certain concepts, rather than deleting the main content. The challenge of concept erasure, as we observe, lies in balancing the thoroughness of target concept erasure with the integrity of non-erasable content. It is crucial to ensure that target concepts are precisely and completely removed from the outputs, while preserving the overall style, the semantic consistency of non-erasable main concepts, and the main content itself.

Based on the above challenges, we first propose the Non-Erasable Features Protector (NEFP), a novel strategy aimed at enhancing the global feature consistency of concept erasure outputs. The core of NEFP is to use orthogonal projection decomposition to recover the global feature information lost in the mapping process of the target concept matrix in previous closed-form methods [3, 7, 9, 23]. Through our proposed anchor fusion mechanism, the enhanced anchor concept matrix is applied to model weight updates to prevent style shifts and preserve additional information except for the specifically targeted features for erasure.

Another key method of our proposed ICE is Augmented Invariant Constraints (AIC), which aims to guide the concept erasure process precisely by introducing invariance constraints while effectively retaining critical semantic information in the representation.

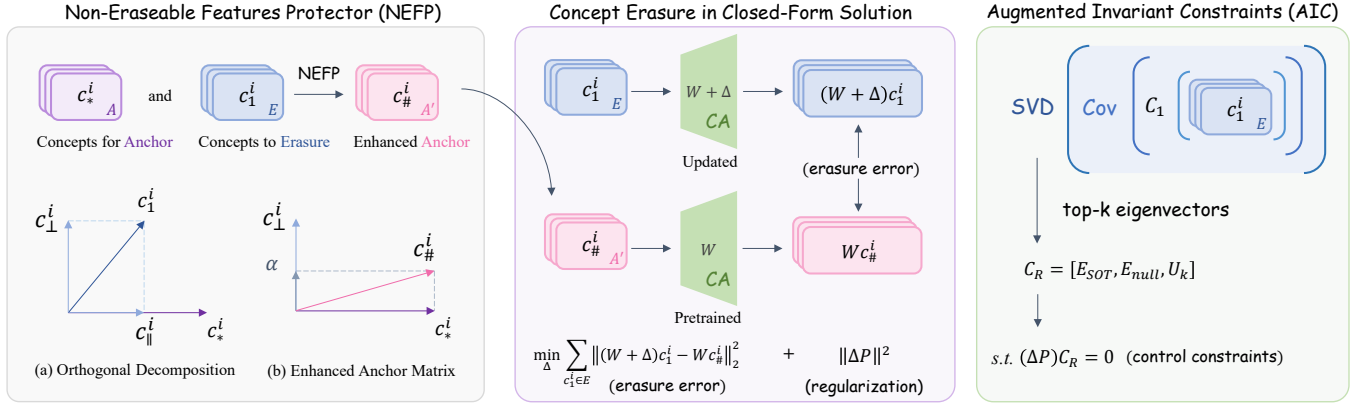


Figure 3: Overview of the proposed ICE. ICE adopts concept erasure in closed-form solutions as the main architecture. The ICE framework introduces two core designs: (1) Non-erasable features protector. By projecting onto orthogonal anchor points through spatial decomposition, it obtains an anchor matrix that retains more information, thus preserving global features during the erasure process. (2) Augmented invariant constraints. Using top-k eigenvectors to obtain invariant constraints, it formulates an optimization problem with Lagrange multipliers during the concept erasure process, derives a closed-form solution, and ensures the retention of key semantics during erasure.

Specifically, our AIC achieves control constraints by introducing key features representing the core semantic of non-erasure targets in the embedded space, which is elaborated in detail in Section 3.3. The optimization problem is finally solved by the Lagrange multiplier method to minimize the erasure error of the target concept while satisfying invariance constraints. Ensuring that while accurately erasing the target concept, the semantic consistency of non-target content can be maintained to the maximum extent.

Overall, our contributions are summarized as follows:

- By re-examining the common problems of concept erasure, we propose a novel concept erasure architecture named ICE, which balances the relationship between target concept erasure and the protection of non-erasable content.
- We propose a novel method NEFP, which retrieves lost global feature information through orthogonal projection decomposition, enhances the coherence of global features before and after erasure, and prevents style shift.
- By designing AIC, we introduce key features in the embedding space to construct control constraints and optimize the erasure process using the method of Lagrange multipliers, ensuring precise erasure of target concepts while maintaining the semantic consistency of non-erasable content.

2 RELATED WORK

2.1 Concept Erasure

In T2I models, concept erasure is crucial for mitigating ethical and legal risks by preventing the generation of specific concepts [6, 20, 45]. Existing research has primarily explored four technical approaches to constrain unintended generation in text-to-image models: (1) posterior image filtering [29, 33], (2) guided reasoning inference [1, 37], (3) selective data retraining [28, 34], and (4) parameter fine-tuning strategies [6, 7, 11, 18, 20, 27, 45]. The first two methods are essentially post-generation correction mechanisms and fail to address the root causes of improper content generation [39]. Therefore,

they cannot fundamentally solve the problem. The third approach, although theoretically sound, faces significant practical limitations. For instance, retraining the mainstream Stable Diffusion model requires more than 150,000 A100 GPU hours of computational resources [35], severely limiting its feasibility. Parameter fine-tuning strategies based on pre-trained models show greater practical value. Model fine-tuning methods [20, 24, 26] preserve prior knowledge by adding a regularization term during training and fine-tuning the T2I diffusion models. This process involves modifying model parameters, typically requiring additional parameter storage and training overhead. ESD [6] eliminates the need for post-inference filtering by directly erasing concepts from model weights. In contrast, we adopt a strategy that leverages closed-form solutions to adjust model parameters directly, eliminating the need for extensive retraining or fine-tuning.

2.2 Closed-Form Solution and Null-Space Constraints

Despite the encouraging progress in concept erasure, the extensive iterations required for parameter fine-tuning can lead to diminished generative capabilities and significant consumption of computational resources. Thus, works that directly modify model parameters without fine-tuning using closed-form solutions have been proposed [3, 7, 9]. UCE [7] proposed a unified method for editing the projection of text-to-image to solve multi-task architectures by directly optimizing the attention matrix of the cross-attention layer and mapping concept keywords to unrelated concepts, thereby achieving rapid erasure. On this basis, the RECE [9] method proposed closed-form editing to only modify the key and value projection matrices in the cross-attention layer and introduced a regularization term to limit the deviation of model parameters to maintain generative capabilities. Moreover, reliance on weighted least squares optimization [3, 7, 9] inherently sets a non-zero lower bound on preservation error. AdaVD [43] firstly implements concept erasure using classical linear algebra orthogonal

complement operations and an adaptive erasure shift design. Building on this, SPEED [23] proposes to use null-space [5, 19, 40, 41] constraints for concept erasure, defining the “null-space” as the subspace that does not alter the feature representations of non-target concepts, thereby reducing the difficulty of optimization. However, erasing the target concepts without carefully protecting overall features and preserving the other primary attributes may introduce global style shift, concept drift, or object absence. In contrast, our proposed ICE addresses these issues by carefully designing the non-erasable features protector mechanism and augmenting the invariant constraints.

3 METHODS

We aim to achieve precise concept erasure through a closed-form solution (Sec. 3.1). To this end, we propose novel non-erasable features protector (NEFP), which constructs an enhanced anchor concept matrix, effectively preventing style shifts while preserving non-target semantic information beyond the specified erasure concepts (Sec. 3.2). Furthermore, we derive augmented invariant constraints (AIC) to maintain consistency in the erased concepts, safeguarding critical information in the representation (Sec. 3.3).

3.1 Concept Erasure in Closed-Form Solution

In T2I diffusion models, each concept is represented by a series of text tokens processed through CLIP [32]. These tokens are subsequently combined into a single concept embedding $c \in \mathbb{R}^d$. In the context of concept erasure, there are two types of concepts with different purposes: the erasure set $E = \{c_1^{(i)}\}_{i=1}^{N_E}$, which consists of N_E target concepts to be removed, and the retained set $R = \{c_0^{(i)}\}_{i=1}^{N_R}$ includes N_R non-target concepts that should be preserved. Building on this, for each target concept $c_1^{(i)}$, an anchor concept $c_*^{(i)}$ is derived to encapsulate the general semantic information of the target concept while excluding its specific attributes.

To derive a closed-form solution for concept erasure, existing methods [7, 23] aim to update the model parameters $\mathbf{W}$ to a new set of weights $\mathbf{W}'$, such that the target concepts are effectively erased while preserving the non-target concepts. For example, UCE [7] achieves concept erasure by formulating it as a weighted least squares optimization problem:

$$\begin{aligned} & \min_{\mathbf{W}'} \sum_{c_1 \in E} \|\mathbf{W}'c_1 - \mathbf{W}c_*\|_2^2 + \sum_{c_0 \in R} \|\mathbf{W}'c_0 - \mathbf{W}c_0\|_2^2 \\ &= \min_{\Delta} \underbrace{\sum_{c_1 \in E} \|(\mathbf{W} + \Delta)c_1 - \mathbf{W}c_*\|_2^2}_{\text{erasure error}} + \underbrace{\sum_{c_0 \in R} \|\Delta c_0\|_2^2}_{\text{preservation error}}, \quad (1) \end{aligned}$$

where the update amount Δ , the change between $\mathbf{W}$ and $\mathbf{W}'$, adjusts the model parameters. By minimizing the erasure error for concepts in the erasure set E , and the preservation error for concepts in the retain set R , the method ensures that target concepts are edited as desired while retaining the integrity of non-target concepts.

The concept embeddings from the erasure set E and the retain set R can be arranged into two structured matrices: $C_1 \in \mathbb{R}^{d \times N_E}$, and $C_0 \in \mathbb{R}^{d \times N_R}$, corresponding to the stacked embeddings of the target and non-target concepts, respectively. For each target concept in C_1 , an anchor concept matrix $C_* \in \mathbb{R}^{d \times N_E}$ is derived. We adopt

Table 1: Cosine similarity with ‘Cartoon Dog’ embedding.

Concept	C_*	$C_\# (\alpha = 0.1)$	$C_\# (\alpha = 0.15)$	$C_\# (\alpha = 0.2)$	C_1
‘Cartoon Dog’	0.3477	0.3681	0.3777	0.3819	0.4037

a null-space constraint method to mitigate distortion on non-target knowledge caused by accumulated preservation error as the number of target concepts increases. Inspired by SPEED [23], we compute the null space of C_0 and construct a projection matrix $\mathbf{P}$ that maps the update Δ onto this null space. This ensures that parameter updates do not interfere with non-target concepts and effectively eliminate preservation errors. To further refine this process, in our paper, an optimization objective is introduced, integrating a regularization term and control constraints, corresponding respectively to minimizing parameter adjustments and ensuring that the updates preserve the target knowledge:

$$\min_{\Delta} \underbrace{\|(\mathbf{W} + \Delta)\mathbf{C}_1 - \mathbf{W}C_\#\|_2^2}_{\text{erasure error}} + \underbrace{\|\Delta\mathbf{P}\|_2^2}_{\text{regularization}}, \text{ s.t., } \underbrace{(\Delta\mathbf{P})C_R = 0}_{\text{control constraints}}. \quad (2)$$

Notably, the matrices $C_\#$ and C_R differ from those used in previous methods, with $C_\#$ derived from our proposed non-erasable features protector and C_R used to augmented invariant constraints. Detailed introductions for $C_\#$ and C_R are provided in the following subsections (Sec. 3.2, 3.3).

After obtaining the solution Δ for Eq.(2) using the Lagrange Multiplier method, the updated weights can be expressed as:

$$\mathbf{W}' \leftarrow \mathbf{W} + \Delta\mathbf{P}, \quad (3)$$

which reflects the updated parameters from the pre-trained model.

3.2 Non-Erasable Features Protector (NEFP)

In the context of concept erasure, prior works with closed-form solutions typically transform a specific target concept $c_1^{(i)}$ (e.g., *Snoopy*) into a more generic and less specific anchor concept $c_*^{(i)}$ (e.g., *Dog*), retaining general semantic information while excluding specific details of the original concept. However, such methods often fail to account for additional information beyond the specifically targeted features for erasure, such as the global style or contextual details. As shown in Figure 2(a), while erasing the *Snoopy* features, the standard method does not produce a *Cartoon Dog* in a comic image with the same global style as the original image, but instead results in a real-world photo of a *Dog*.

To address this limitation, we propose the Non-Erasable Features Protector (NEFP) mechanism, designed to enhance the existing anchor concept matrix $C_*^{(i)}$. In prior work, it is assumed that the obtained $c_*^{(i)}$ contains the original concept with the specified features erased. However, during the mapping from $c_1^{(i)}$ to $c_*^{(i)}$, other information, such as style and additional details, may be unintentionally lost. To mitigate this, we project C_1 onto a subspace orthogonal to C_* , thereby retaining the detailed information that might have been omitted in C_* . The orthogonal component is computed as follows:

$$\begin{aligned} C_\perp &= C_1 - \text{Proj}_{C_*}(C_1) = C_1 - \frac{C_1 \cdot C_*^\top}{\|C_*\|^2} \cdot C_* \\ &= C_1 - \lambda_{\text{proj}} \cdot C_*, \end{aligned} \quad (4)$$

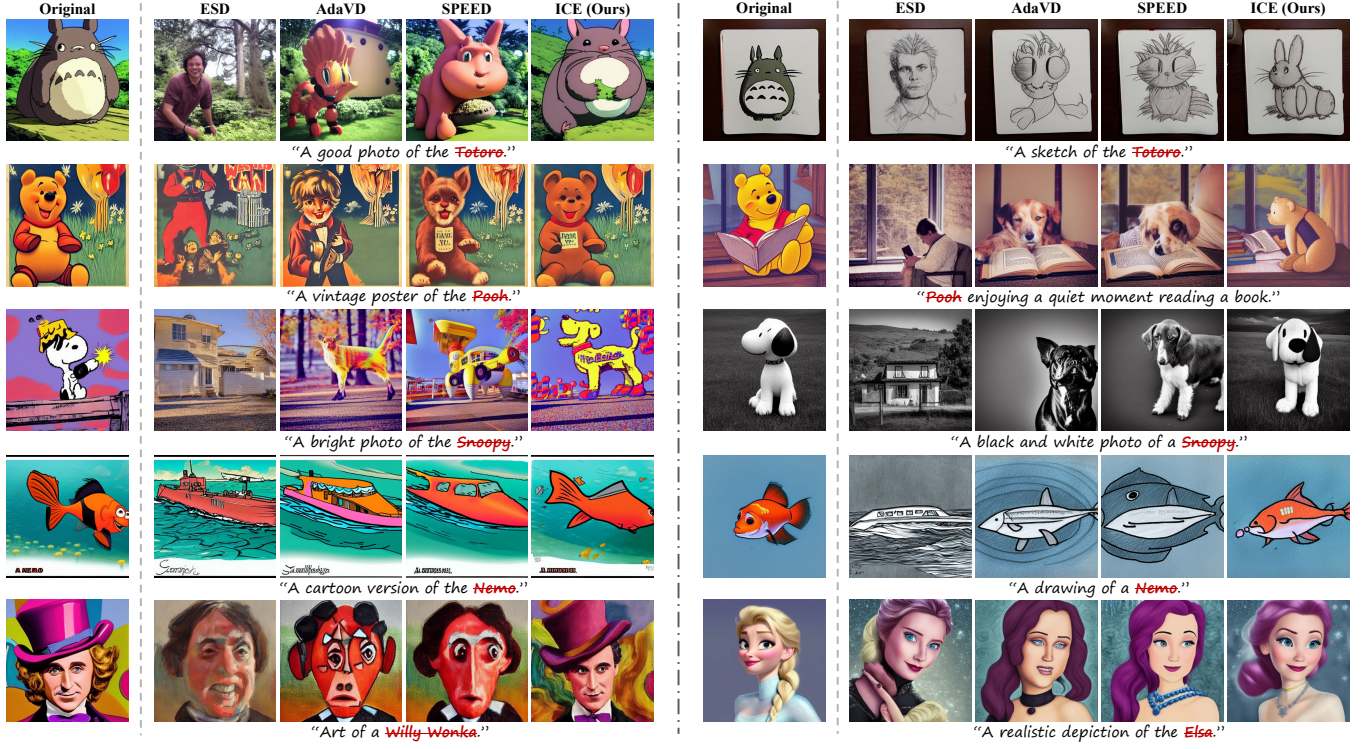


Figure 4: Qualitative comparison. We compare our method with ESD [6], AdaVD [43], and SPEED [23] under various prompts in multiple scenarios such as cartoon and movie characters. Our method achieves precise feature erasure, i.e., ensuring both effective erasure while better maintaining global style and attribute consistency.

where $C_{\perp}$ represents the portion of C_1 that is independent of C_* . This ensures that additional details not captured in C_* are effectively preserved. The projection coefficient is given by:

$$\lambda_{\text{proj}} = \frac{C_1 \cdot C_*^T}{\|C_*\|^2}, \quad (5)$$

which quantifies the degree to which C_1 aligns with C_* . In particular, it measures the magnitude of the component of C_1 projected onto the direction defined by C_* . To integrate the preserved orthogonal component $C_{\perp}$, we propose a fusion mechanism that incorporates the additional details retained in $C_{\perp}$ into the anchor concept matrix C_* . The enhanced matrix $C_{\#}$ is defined as:

$$\begin{aligned} C_{\#} &= C_* + \alpha \cdot C_{\perp} = C_* + \alpha \cdot \left(C_1 - \frac{C_1 \cdot C_*^T}{\|C_*\|^2} \cdot C_* \right) \\ &= (1 - \alpha \lambda_{\text{proj}}) \cdot C_* + \alpha \cdot C_1. \end{aligned} \quad (6)$$

Where α is a scalar parameter that controls the degree to which the orthogonal component $C_{\perp}$ contributes to the enhanced matrix $C_{\#}$. A smaller α prioritizes the original information in C_* , while a larger α allows more detailed information from $C_{\perp}$ (e.g., style-related features) to be incorporated. By adjusting α , we can effectively balance the original information and the additional details embedded in $C_{\perp}$. We conducted a simple experiment on the ‘Snoopy’ example mentioned at the beginning of this subsection, calculating the cosine similarity between different matrices and the embedding of ‘Cartoon Dog’. The results in Table 1 show that compared to C_* , the adoption of $C_{\#}$ improves the retention of global style.

From above experiment and observation, embedding the preserved orthogonal component $C_{\perp}$ into the anchor concept C_* allows the model to retain crucial attributes, such as the original style and background, while effectively removing the specific features targeted for erasure. The above analysis focuses on instance erasure. In contrast, for style erasure, we select a negative α tailored to the characteristics of this task to achieve better erasure performance.

3.3 Augmented Invariant Constraints (AIC)

In parallel, we introduce invariants into the embedding space under control constraints to guide the erasure process while preserving the critical semantics of the representation. This approach aims to mitigate the challenge of maintaining consistency in erased concepts, as traditional erasure methods may sometimes result in the erased concept transforming into an unrelated object (e.g., *Snoopy becoming a sailboat*, as shown in Figure 2(b)) or the original object disappearing entirely from the representation (see Figure 2(c)).

In prior methods [23], the CLIP-encoded [SOT] token (a fixed prefix token added during tokenization to ensure consistent embeddings), consistent across prompts, and the null-text embedding, representing an unconditional generation, are stacked together to serve as stable references unaffected by variations in prompts. Building on this, we stack the essential features representing the core semantics of non-erasure targets with these invariants to enforce control constraints. To preserve the primary semantic information without additional manually provided textual prompts, we process the target concept matrix C_1 by computing its covariance matrix Σ :

Table 2: Quantitative comparison. Our ICE achieves best results in both instance erasure and style erasure, standing out particularly in the detailed subjective evaluations of the user study.

	Instance Erasure		Style Erasure		User Study			Time Cost (s)	
Metrics	CER ↑	DINO ↑	CER ↑	CLIP-T ↑	Style Consistency ↑	Attribute Retention ↑	Overall Performance ↑	Train / Edit ↓	Inference ↓
ESD (SDv1.4)	0.338	0.671	0.305	0.239	3.750%	6.875%	5.000%	363	2.6
AdaVD (SDv1.4)	<u>0.363</u>	0.807	<u>0.319</u>	0.248	0.625%	3.125%	1.875%	/	1
SPEED (SDv1.4)	0.351	0.803	0.314	<u>0.268</u>	<u>2.500%</u>	<u>16.875%</u>	<u>6.875%</u>	1	0.3
SPEED (SDXL)	0.341	0.659	0.307	0.238	/	/	/	77	5
Our ICE (SDv1.4)	0.371	0.873	0.322	0.271	93.125%	73.125%	86.250%	1	0.34
Our ICE (SDXL)	0.35	<u>0.864</u>	0.313	0.246	/	/	/	77	5

$$\Sigma = \text{Cov}(\mathbf{C}_1) = \frac{1}{N_E - 1} (\mathbf{C}_1 - \bar{\mathbf{C}}_1) (\mathbf{C}_1 - \bar{\mathbf{C}}_1)^\top, \quad (7)$$

where $\bar{\mathbf{C}}_1$ is the mean of the target concepts matrix:

$$\bar{\mathbf{C}}_1 = \frac{1}{N_E} \mathbf{C}_1 \mathbf{1}_{N_E}^\top. \quad (8)$$

By applying Singular Value Decomposition (SVD) to the covariance matrix Σ , we decompose it into three components:

$$\text{SVD}(\Sigma) = \{\mathbf{U}, \mathbf{\Lambda}_\Sigma, \mathbf{U}^\top\}, \quad (9)$$

where the matrix of eigenvectors $\mathbf{U}$ represents the principal directions of variability in the target concepts. To focus on the most significant features, we select the top- k eigenvectors from $\mathbf{U}$, denoted as $\mathbf{U}_k = \mathbf{U}[:, :k]$, where k is a hyperparameter set to 5. The choice is based on the assumption that the top principal components capture the majority of the meaningful variation in the target concepts, thus representing their critical semantic features. Finally, we integrate these components into a unified representation:

$$\mathbf{C}_R = [\mathbf{E}_{\text{SOT}}, \mathbf{E}_{\text{null}}, \mathbf{U}_k], \quad (10)$$

thereby obtaining $\mathbf{C}_R$, which acts as a constraint term in the optimization objective to regulate the erasure process. Compared to previous method [23], we novelly incorporate the critical semantic features of the target as part of the constraints, ensuring invariants guide the erasure process effectively while preserving semantics and retaining essential information in the representation.

Since the matrices $\mathbf{C}_\#$ and $\mathbf{C}_R$ have been obtained, we next formulate the function to solve Eq.2 using the Lagrange Multiplier method. The objective is to minimize the following constrained optimization problem:

$$\mathcal{L}(\Delta, \Lambda) = \|(\mathbf{W} + \Delta\mathbf{P})\mathbf{C}_1 - \mathbf{W}\mathbf{C}_\#\|^2 + \|\Delta\mathbf{P}\|^2 + \Lambda^\top (\Delta\mathbf{P})\mathbf{C}_R, \quad (11)$$

where Λ serves as the Lagrange multiplier. To solve this optimization problem, we derive the following conditions by setting the partial derivatives of $\mathcal{L}(\Delta, \Lambda)$ with respect to Δ and Λ to zero:

$$\begin{cases} \frac{\partial \mathcal{L}(\Delta, \Lambda)}{\partial \Delta} = 0, \\ \frac{\partial \mathcal{L}(\Delta, \Lambda)}{\partial \Lambda} = (\Delta\mathbf{P})\mathbf{C}_R = 0. \end{cases} \quad (12)$$

Following the derivation in Supplementary Material B, we arrive at the closed-form solution of Eq.2, which is expressed as:

$$(\Delta\mathbf{P})_{\text{ICE}} = \mathbf{W} \left((1 - \alpha\lambda_{\text{proj}})\mathbf{C}_\#\mathbf{C}_1^\top - (1 - \alpha)\mathbf{C}_1\mathbf{C}_1^\top \right) \mathbf{P}\mathbf{Q}\mathbf{M}, \quad (13)$$

where

$$\mathbf{M} = (\mathbf{C}_1\mathbf{C}_1^\top \mathbf{P} + \mathbf{I})^{-1}, \quad \mathbf{Q} = \mathbf{I} - \mathbf{M}\mathbf{C}_R (\mathbf{C}_R^\top \mathbf{P}\mathbf{M}\mathbf{C}_R)^{-1} \mathbf{C}_R^\top \mathbf{P}. \quad (14)$$



Figure 5: Style erase comparison. Our ICE can better balance style erasure while preserving semantic consistency.

The resulting $\Delta\mathbf{P}$ updates model parameters per Eq. 3.

4 EXPERIMENTS

4.1 Qualitative Comparison

For qualitative comparison, we mainly compare our method with three methods, including ESD [6], AdaVD [43], and SPEED [23]. To ensure a fairer comparison, we implemented all methods using SDv1.4 [35]. All compared methods are implemented using their default configurations from the corresponding official repositories.

Instance Erase Comparison. We select a variety of famous cartoon instances and movie character instances and evaluate various methods across diverse scenarios, ranging from image generation in the vintage poster style to the creation of realistic depictions. The experimental results are shown in Figure 4. We find that ESD erases the target object but also removes key content or generates irrelevant instances. AdaVD and SPEED cause semantic shifts (e.g., replacing “Nemo” with a ship) and fail to preserve the original style (e.g., turning “Snoopy” into a real-world dog). In contrast, our method ensures precise concept erasure, making the generated image unrecognizable as the target object while preserving original attributes. Additionally, it achieves an editing-like effect, maintaining better consistency in unchanged elements, such as the background, between the generated images and those from the original model. Meanwhile, we compared our method with SPEED on the SDXL model [31], and the results can be found in Supplementary Material E.

Style Erase Comparison. As illustrated in Figure 5, taking the removal of the “Van Gogh” and “Expressionism” styles as examples, the images generated by ESD exhibit significant semantic differences compared to the original ones. The AdaVD and SPEED methods are not effective enough in erasing the style, as some texture information from the original style remains in the portrait, and

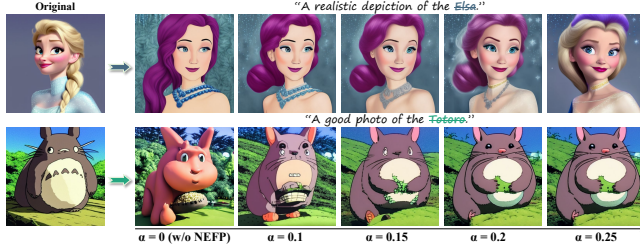


Figure 6: Ablation study on NEFP module and α for instance erasure task. Our proposed NEFP effectively enhances global style consistency. We select $\alpha = 0.2$ to strike a balance between erasure effectiveness and style consistency preservation.

certain features, such as the eyes, are not naturally transformed. Compared to other methods, our ICE achieves a better balance between erasing the original style and maintaining the semantic consistency of the images generated by the original model.

4.2 Quantitative Comparison

We conduct experiments on instance erasure and artistic style erasure to assess the concept erasure performance compared to baseline methods [6, 23, 43], using an NVIDIA A100 GPU. Conventional concept erasure methods typically utilize the CLIP Text Score [32], which measures the similarity between text and image, to evaluate the erasure effect. However, this metric mainly evaluates the target’s removability without considering that the purpose of concept erasure is to prevent the model from generating specific concepts, rather than deleting the main content or generating unrelated instances. For instance, generating a completely black image might score highly on this metric due to its significant contrast with the target, but such a result is far from ideal. To evaluate the erasure effect more comprehensively, we propose a new metric, the CLIP Erasure Ratio (CER), which considers the effectiveness of erasure, the prevention of random concept drift, and the preservation of main content. Specifically, CER is defined as the ratio of the CLIP Image Score (CIS) to the CLIP Text Score (CTS), providing a balanced measure of these factors: $CER \uparrow = \frac{CIS \uparrow}{CTS \uparrow}$. The task setting requires the generated instances after erasure to be difficult to recognize as the original instances (smaller CTS is better) while presenting alternative expressions related to the erasure target (larger CIS is better). Thus, a higher CER indicates a more effective erasure with minimal sacrifice of non-erased content.

In addition, for each task, we utilize specific metrics to further assess performance. For instance erasure, we use the DINO score [12] to measure the distance between the erased and original image pairs, assessing the style consistency performance within the set in alignment with the key goal of our task. While for style erasure, we employ CLIP-T to assess the alignment between the erased image and the text (excluding the erased style word), which allows us to evaluate how well the content is preserved. Meanwhile, the time cost is measured to evaluate the efficiency of each method. The experimental results are shown in columns 2-5 and 9-10 of Table 2, where our proposed ICE achieves best scores in all metrics for both instance erasure and style erasure tasks, while maintaining efficient generation.

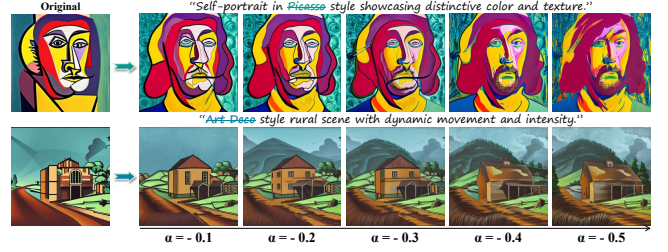


Figure 7: Ablation study on α of NEFP module for style erasure task. A larger absolute value of α enhances style erasure, while a smaller one preserves consistency.

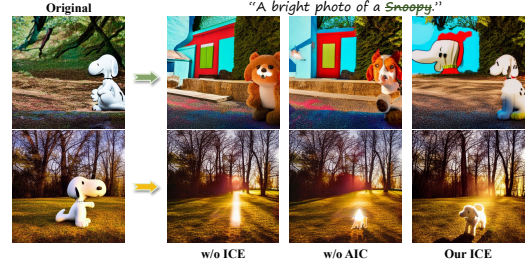


Figure 8: Ablation study on AIC module. Our proposed AIC effectively addresses the issues of concept drift and content absence, achieving better attribute retention.

4.3 User Study

In addition to the metric evaluation, we conduct a user study to assess the results from human perspectives. Sixteen experts in the field of vision and graphics participate in it, completing 10 sets of comparison tasks involving 40 images. Participants are asked to select the best concept-erased image from three critical dimensions: style consistency, attribute retention, and overall performance. In total, we obtain 480 votes, and the voting proportions are shown in columns 6-8 of Table 2. Our proposed ICE achieves outstanding scores of 93.12%, 73.12%, and 86.25% in the three aspects mentioned above, respectively. Additionally, participants complete another 5 sets of rating tasks, providing ratings across more detailed evaluation metrics, yielding a total of 1,600 rating records. For more details on the user study and visualizations of the results, please refer to Supplementary Material A.

4.4 Ablation Study

To validate the effectiveness of our method, we conduct the following ablation experiments on the two proposed modules, NEFP (Sec. 3.2) and AIC (Sec. 3.3), as well as the hyperparameter involved. **Ablation Study on NEFP.** To assess the effectiveness of the NEFP module and the impact of the scalar parameter α , we present the outputs of our method for different α values in instance and style erasure. Considering the varying focuses of each task, we use a positive α for instance erasure and a negative α for style erasure.

As shown in Figure 6, when $\alpha = 0$ (i.e., w/o NEFP), the images obtained after instance erasure suffer from global style shifts. For example, the originally generated “Elsa” by SDv1.4, which has a three-dimensional style, shifts to a flat illustration style after erasure. With the NEFP module, an increase in α leads to the resulting images retaining the style more consistent with those generated by the

Table 3: Quantitative ablation on the proposed components NEFP and AIC on instance erasure task.

Methods	CER $\uparrow$	DINO $\uparrow$
Our w/o AIC	0.366	0.861
Our w/o NEFP	0.367	0.815
Our ICE	0.371	0.873

original model. To balance feature erasure and style preservation, we set $\alpha = 0.2$ for our NEFP module in the instance erasure task.

For style erasure, as shown in Figure 7, with α set to a negative value, a larger absolute value leads to a more significant style erasure ability in the edited model. In the case of Art Deco style erasure, the left example retains stronger semantic consistency with minimal style changes, while the right example shows how the model transforms the structured style into a more refined, softer appearance. We set $\alpha = -0.4$ to balance the erasure effectiveness and semantic consistency in the style erasure task.

Ablation Study on AIC. The ablation visual results under the AIC module are shown in Figure 8, where we present the classic “Snoopy” erasure task as an example. Without the introduction of our proposed ICE module, the output images suffer from issues such as concept drift and the absence of main content. With the NEFP module present, the generated images show some improvement but fail to resolve the issues above completely. By incorporating the AIC module, issues such as object disappearance or attribute misalignment are resolved, aligning with our goal of preserving the semantic consistency and the main content. The quantitative results of the ablation experiment in the instance erasure task are shown in Table 3, demonstrating the effectiveness of our proposed components NEFP and AIC. Notably, the NEFP module significantly improves the DINO metric, which aligns with its motivation to maintain a consistent global style.

4.5 More applications.

To further demonstrate the versatile application of our ICE in a variety of generative tasks, we conduct the following additional experiments. As shown in Fig 9(a), our method can erase specific artistic styles and character concepts while retaining other content information in the image. This function is significant for copyright protection, allowing image modifications without infringing on copyrights. (2) Exposure concept erasure in image semantics. ICE can effectively erase exposure concepts in image semantics, as shown in Fig 9(b). Even without explicitly specifying the specific content to be erased, ICE can intelligently add clothing to key areas. (3) Attribute suppression. ICE can also be extended to the field of attribute suppression. For example, as shown in Fig 9(c), we demonstrate the suppression of the color attribute “red” and the material attribute “glass.” Fig 9(c) indicates that ICE can decouple strongly coupled concepts, allowing users to make flexible modifications and adjustments to specific attributes.

5 CONCLUSIONS

In this paper, the proposed ICE method addresses the key issues of model performance degradation and over-erasure caused by previous concept erasure methods by innovatively constructing an



Figure 9: More applications. We extend ICE to versatile generation tasks, including (a) Artistic and character erasure. (b) Nude erasure. (c) Attribute suppression, highlighting its substantial potential for diverse applications.

orthogonal anchor concept matrix and designing a dynamic invariance constraint mechanism. This marks an important step towards maintaining ethical and socially responsible T2I generation models. The ICE framework includes two core designs: (1) Based on parameter subspace decomposition, an orthogonal anchor projection is introduced. Establishing a high-dimensional orthogonal to the target concept retains crucial information during the erasure process. (2) Enhanced invariance constraints with adaptive capabilities are constructed. Using the Lagrange multiplier method, an optimization problem is formulated, and a closed-form solution is derived to ensure the protection of key semantics during erasure.

Limitations and future works. Some instances to be erased do not have closely related replacements with the same attribute, resulting in generated images that, while unrecognizable as the erased target, may produce objects lacking real-world significance. In future research, we plan to develop more robust and versatile intelligent concept-erasing technologies in response to such limitations.

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